

11.17 Classwork: Cumulative Frequency and Normal Models — Markscheme

1. [Maximum mark: 17]

AA SL 2023 Nov P1 TZ2 Q7

(a)(i) Find p . [1]

$p = 60 - 3 - 18 - 30 = 9$ **A1**

(a)(ii) Write down the modal class. [1]

$600 < n \leq 800$ **A1**

Do not accept 700.

(b)(i) Estimate the median number of tickets sold. [1]

The median is the 30th performance. From the cumulative frequency graph, the median is 600 tickets. **A1**

(b)(ii) Estimate the number of performances where at least 80% of the tickets were sold. [3]

80% of 800 is 640. **A1**

From the cumulative frequency graph, about 40 performances sold fewer than 640 tickets. **A1**

Therefore $60 - 40 = 20$ performances sold at least 80% of the tickets. **A1**

(c)(i) State one disadvantage of surveying only the first 5% of the audience as they leave. [1]

Any reasonable answer identifying bias, with a reason. For example: the first people leaving may come from the same section, group, or priority seating area, so the sample may not represent the whole audience. **A1**

(c)(ii) Describe how to collect feedback from 5% of the audience using systematic sampling. [2]

Select people at regular intervals. **A1**

Since $5\% = \frac{1}{20}$, survey every 20th person. **A1**

(c)(iii) State the sampling method which should be used if the survey is to be representative of the number of children and adults. [1]

Quota sampling **A1**

(d)(i) Estimate the number of people who spent between \$3 and \$25. [2]

From the box plot, \$3 is the lower quartile and \$25 is the maximum, so this interval contains 75% of the audience. **M1**

$0.75 \times 36\,000 = 27\,000$ people. **A1**

(d)(ii) Half the audience spent less than $\$a$. Estimate a . [1]

The median amount spent is $\$7$, so $a = 7$. **A1**

(e)(i) Calculate the mean number of tickets expected this year. [3]

Last year the total number of tickets sold was 36 000, so the old mean was

$$\frac{36\,000}{60} = 600.$$

Old mean is 600 tickets. **A1**

Adding 17 tickets to each performance adds 17 to the mean. **M1**

New mean $= 600 + 17 = 617$ tickets. **A1**

Alternative: compute $\frac{36\,000 + 60(17)}{60} = 617$.

(e)(ii) State the effect on the variance. [1]

Adding the same constant to every value has no effect on the variance. **A1**

2. [Maximum mark: 16]

AA SL 2023 Nov P2 TZ1 Q9

(a) Find $P(82.4 < H < 87.3)$. [2]

Recognize that the three probabilities sum to 1. **M1**

$$0.213 + P(82.4 < H < 87.3) + 0.409 = 1$$

$$P(82.4 < H < 87.3) = 0.378$$

A1

(b) Find μ and σ . [5]

Use inverse normal with $P(H < 82.4) = 0.213$ and $P(H < 87.3) = 1 - 0.409 = 0.591$.
M1

$$\mu + z_1\sigma = 82.4, \quad \mu + z_2\sigma = 87.3$$

where $z_1 = \text{invNorm}(0.213) = -0.796055\dots$ and $z_2 = \text{invNorm}(0.591) = 0.230118\dots$
A1A1

Attempt to solve the two equations in μ and σ . **M1**

$$\mu = 86.2011\dots, \quad \sigma = 4.77502\dots$$

$$\mu = 86.2, \quad \sigma = 4.78$$

A1

Accept equivalent GDC setup using normal CDF or inverse normal.

(c)(i) Find the probability that exactly 32 plants are ready to harvest. [2]

Recognition of binomial distribution. **M1**

$$X \sim B(100, 0.409)$$

$$P(X = 32) = 0.0157931\dots = 0.0158$$

A1

(c)(ii) Given that fewer than 44 plants are ready to harvest, find the probability that exactly 32 plants are ready to harvest. [4]

$$P(X < 44) = 0.702975 \dots$$

A1

Recognition of conditional probability in context.

M1

$$P(X = 32 \mid X < 44) = \frac{P(X = 32)}{P(X < 44)}$$

A1

$$\frac{0.0157931 \dots}{0.702975 \dots} = 0.0224661 \dots = 0.0225$$

A1

FT: award marks for correct conditional setup using the candidate's binomial probability from part (c)(i).

(d) Find d .

[3]

Since the IQR is 4.52, the quartiles are

$$Q_1 = 92.8 - 2.26 = 90.54, \quad Q_3 = 92.8 + 2.26 = 95.06.$$

One correct quartile value may be seen.

A1

Use the fact that $P(F < 90.54) = 0.25$ or $P(F < 95.06) = 0.75$.

M1

$$\frac{90.54 - 92.8}{d} = -0.674489 \dots$$

$$d = 3.35068 \dots = 3.35$$

A1

Accept equivalent GDC method solving for d .